

WORKERS, BICOMODULES, AND PARAMETRIC RIGHT ADJOINTS: A CORRESPONDENCE AND A CONJECTURE

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ABSTRACT. Spivak, Garner and Fairbanks proved that bicomodules of polynomial comonads are exactly the parametric right adjoints between the associated copresheaf categories, and that every such functor factors as a connected-limit-preserving (right adjoint) part followed by a coproduct (left adjoint) tail that carries no bicomodule data. Separately, we have proved that the Workers state-composition discipline — the Dirichlet grading $\Delta S \otimes \Delta T = \Delta(S \times T)$ on store comonads — is a canonical *retract*, not a fibre, of the composition grading $\Delta S \triangleright \Delta T$ used by Braithwaite–Hedges–Mihejevs, the idempotent collapsing each branch map to its self-evaluation. This note sets the two pictures side by side. The retract’s “keep versus discard” split has the same shape as the factorization’s “right-adjoint content versus Σ -tail,” which suggests the identification *worker* = *bicomodule/parametric right adjoint*. We formulate this as a precise conjecture and, rather than assert it, we isolate one concrete obstacle — an apparent *reversal of adjointness* between the two splits — that a proof must confront. The correspondence is established literature; the identification is a hypothesis under test.

1. INTRODUCTION

Two research programs describe composition of stateful, interacting systems in the language of polynomial functors, and they have never been made to talk to each other. This note is an attempt to build the bridge — or, more honestly, to survey the ground it must cross and to plant a flag where the hard step is.

The first picture: aggregation and parametric right adjoints. In the polynomial-functor ecosystem [3], a small category is the same thing as a polynomial comonad, and the natural maps *between* such categories come in two flavours: the “vertical” maps are retrofunctors (cofunctors), and the “horizontal” maps are two-sided *bicomodules*. Spivak, Garner and Fairbanks [4] identify these bicomodules concretely: a bicomodule from the comonad c to the comonad d is exactly a *parametric right adjoint* (equivalently a connected-limit-preserving functor, equivalently a data-migration / aggregation functor) between the copresheaf categories $c\text{-}\mathbf{Set} \rightarrow d\text{-}\mathbf{Set}$. Their Proposition 3.6 factors any such functor into a genuinely “polynomial” right-adjoint part and a coproduct tail that is not bicomodule data — the aggregation (Σ) step. This is the setting in which “how does data flow and combine along a functor” receives a clean categorical answer.

The second picture: workers and state. In work aimed at the semantics of stateful agents [5], one models a piece of local state S by the *store comonad*, carried by the polynomial $\Delta S = S \cdot y^S$. Two workers with states S and T can be composed in (at least) two ways. The *Workers* discipline entangles their states via the Dirichlet product, $\Delta S \otimes \Delta T = \Delta(S \times T)$; the Braithwaite–Hedges–Mihejevs (BHM) discipline nests them via functor composition, $\Delta S \triangleright \Delta T$, a store-of-stores. These are genuinely different polynomials — at $|S| = |T| = 2$ they are $4y^4$ and $8y^4$ — and our main structural result about them is that the first is a canonical *retract* of the second (Theorem 4.1, proved in [5]): there are polynomial maps σ and r with $r \circ \sigma = \text{id}$, and the idempotent $e = \sigma \circ r$ collapses each branch map $g: S \rightarrow T$ to the constant at its self-evaluation $g(s)$. The discarded data

— the non-constant branch maps — is exactly the store-of-store structure that nesting sees and entanglement forgets.

The question. Both pictures split a rich construction into “content that is kept” and “a tail that is thrown away.” In the aggregation picture the kept part is the right adjoint and the tail is the coproduct Σ ; in the workers picture the kept part is the entangled state $\Delta(S \times T)$ and the tail is the non-constant branch maps. The shapes match. It is tempting — and it has been repeatedly tempting, across several reading sessions — to conclude that the workers/BHM distinction is *precisely* the parametric-right-adjoint distinction: that a worker is a bicomodule and the retract’s kernel is the Σ -aggregation tail.

The purpose of this note is to state that temptation precisely and then to resist asserting it. We recall the SGF correspondence (§3) and the workers retract (§4) in a common language, formulate the identification as Conjecture 5.1 (§5), and — this is the point — exhibit a specific obstacle to it. The obstacle is an *apparent reversal of adjointness*: the data the retract discards (branch maps $S \rightarrow T$) is *exponential*, hence right-adjoint-flavoured, whereas the data the SGF factorization discards is a *coproduct*, hence left-adjoint-flavoured. If the identification is to hold, this reversal must be explained, not ignored. We record it as a gap for a subsequent proof session, together with a second, type-level gap about *which* two comonads (c, d) the workers construction is a bicomodule between.

Status of the claims. The SGF correspondence is established, peer-reviewed literature [4], deep-read for this note. The workers retract is our own result, proved and machine-checked at small size [5]. The identification of the two is a *conjecture*; every statement in §5 that goes beyond the two established inputs is flagged as such. No new theorem is proved here.

Outline. Section 2 fixes the polynomial-functor conventions, the store comonad, and the two products $\otimes, \triangleright$. Section 3 states the SGF correspondence and its factorization in our own words. Section 4 recalls the workers retract. Section 5 states the conjecture and its two gaps. Section 6 says what a proof session must settle.

2. PRELIMINARIES

We work in the category **Poly** of polynomial functors $\mathbf{Set} \rightarrow \mathbf{Set}$, following the conventions of [3].

Definition 2.1 (polynomials and their maps). A *polynomial* is a coproduct of representables $p = \sum_{s \in S_p} y^{p[s]}$, acting on sets by $p(X) = \sum_{s \in S_p} X^{p[s]}$; we call S_p its *shapes* (or positions) and $p[s]$ its set of *directions* at s . A morphism $f: p \rightarrow q$ is a pair $f = (f_1, f^\#)$ with $f_1: S_p \rightarrow S_q$ on shapes and $f_s^\#: q[f_1 s] \rightarrow p[s]$ *backward* on directions. Composition is $(g \circ f)_1 = g_1 f_1$ and $(g \circ f)_s^\# = f_s^\# \circ g_{f_1 s}^\#$: backward maps compose in reverse. We track this reversal explicitly; it is where naïve shape-only reasoning fails.

Poly carries two monoidal structures we need.

Definition 2.2 (the two products). The *composition product* is $(p \triangleright q)(X) = p(q(X))$, with unit y ; it is strictly associative. The *Dirichlet product* $p \otimes q$ has shapes $S_p \times S_q$ and directions $p[s] \times q[t]$ at (s, t) , with unit y .

Definition 2.3 (polynomial comonads are categories). A comonoid for $(\mathbf{Poly}, \triangleright, y)$ is the same data as a small category [1, 3]: shapes are objects, directions at s are the morphisms out of s , the counit picks out identities and the comultiplication composes. Under this dictionary the framed bicategory of polynomial comonads is written $\mathbf{Cat}^\#$ [3]. A polynomial comonad c presents a category we also call c , with copresheaf category $c\text{-}\mathbf{Set}$.

Definition 2.4 (store comonad). For a set S , the *store* (or *costate*) polynomial is $\Delta S := S \cdot y^S$ — shapes S , with $\Delta S[s] = S$ at every shape. It carries the store comonad $\Delta S(X) = S \times X^S$, whose comultiplication is $\delta_1(s) = (s, \text{id}_S)$, $\delta_s^\sharp(i, j) = j$, and whose counit is $\varepsilon_1(s) = *$, $\varepsilon_s^\sharp(*) = s$. As a category ΔS is the *codiscrete* (indiscrete) category on S : exactly one morphism between any two objects.

Finally, the object of study on the SGF side.

Definition 2.5 (parametric right adjoint). Let C have a terminal object 1 . A functor $F: C \rightarrow D$ is a *parametric right adjoint* (PRA), or *prafunctor*, if the first factor of the canonical factorization

$$C \xrightarrow{F/1} D/F1 \xrightarrow{U} D$$

is a right adjoint, where $F/1$ sends c to $Fc \rightarrow F1$ and U forgets down to D [4, Def. 3.5]. For endofunctors of **Set** this is equivalent to preserving connected limits, and to being polynomial [4, Prop. 2.5]. A *bicomodule* from a polynomial comonad c to a polynomial comonad d is a polynomial equipped with compatible left c - and right d -coactions; it is the “horizontal” notion of map between the categories c and d .

3. THE SPIVAK–GARNER–FAIRBANKS CORRESPONDENCE

We restate the two results of [4] that this note leans on, in our own words. Both are established; we claim no part of them.

Theorem 3.1 ([4, Thm. 4.28]). *The framed bicategories $\mathbf{Cat}^\sharp$ and $\mathbf{Comod}(\mathbf{Poly})$ are equivalent. Under the equivalence: objects are small categories, i.e. polynomial comonads; the horizontal (loose) maps from c to d are the (c, d) -bicomodules, which are exactly the prafunctors $c\text{-}\mathbf{Set} \rightarrow d\text{-}\mathbf{Set}$; the vertical (tight) maps are the comonoid homomorphisms, i.e. retrofunctors (not ordinary functors); and the 2-cells are natural transformations.*

The identification of the horizontal maps is made pointwise by a chain of six equivalent descriptions.

Proposition 3.2 ([4, Prop. 3.9]). *For polynomial comonads c, d there are equivalences*

$$\mathbf{Comod}(\mathbf{Poly})(c, d) \simeq \mathbf{PRA}(c\text{-}\mathbf{Set}, d\text{-}\mathbf{Set}) = \mathbf{CLP}(c\text{-}\mathbf{Set}, d\text{-}\mathbf{Set}) \simeq [d, \mathbf{Set}[c]] \simeq [d, \mathbf{Fam}((c\text{-}\mathbf{Set})^{\text{op}})],$$

where $\mathbf{CLP}$ denotes the connected-limit-preserving functors.

The result we actually want to compare against workers is the factorization of a single prafunctor, because it is what performs the “keep versus discard” split.

Proposition 3.3 ([4, Prop. 3.6], aggregation factorization). *Every prafunctor $F: c\text{-}\mathbf{Set} \rightarrow \mathbf{Set}$ factors as*

$$c\text{-}\mathbf{Set} \xrightarrow{F/1} \mathbf{Set}^{F1} \xrightarrow{\Sigma} \mathbf{Set},$$

where $F/1$ is a right adjoint — the genuine polynomial/bicomodule content — and Σ is the coproduct, a left adjoint “aggregation tail” that carries no bicomodule data.

The moral of Proposition 3.3 is exactly a keep/discard split: the bicomodule content lives in the right-adjoint factor $F/1$, and the coproduct Σ is a tail one adds on top and which the correspondence does not see.

Remark 3.4 (handedness). Variance conventions differ across the literature: [4] writes bicomodules as $c\text{-}\mathbf{Set} \rightarrow d\text{-}\mathbf{Set}$, while Spivak’s “Categories by Kan extension” writes the reversed $c'\text{-}\mathbf{Set} \rightarrow c\text{-}\mathbf{Set}$ via the opposite handedness on bicomodules. It is the same theorem, but any concrete identification must first pin the convention. We use the [4] direction throughout.

4. THE WORKERS RETRACT

We recall the result of [5]. It is proved there in full and verified computationally for shape sets of size ≤ 3 ; we import it and reuse its notation.

Write $A := \Delta S \otimes \Delta T = \Delta(S \times T)$ (the Workers grading) and $B := \Delta S \triangleright \Delta T$ (the BHM grading). By Definition 2.4 and the two product formulas, A has shapes $S \times T$ with directions $S \times T$ everywhere, while B has shapes $\{(s, g) : s \in S, g : S \rightarrow T\} = S \times T^S$ and directions $S \times T$ everywhere. So a shape of B carries a whole *branch map* $g : S \rightarrow T$, where a shape of A carries only a point $t \in T$.

Theorem 4.1 ([5, Thm. 1.1]). *Define $\sigma : A \rightarrow B$ by $\sigma_1(s, t) = (s, \text{const}_t)$ with identity backward map, and $r : B \rightarrow A$ by $r_1(s, g) = (s, g(s))$ (self-evaluation) with identity backward map. Then $r \circ \sigma = \text{id}_A$, while $\sigma \circ r \neq \text{id}_B$ whenever $|S|, |T| \geq 2$. Hence $A = \Delta S \otimes \Delta T$ is a non-trivial retract of $B = \Delta S \triangleright \Delta T$.*

Corollary 4.2 (the discarded data). *The idempotent $e := \sigma \circ r$ has $e_1(s, g) = (s, \text{const}_{g(s)})$ and identity backward map. It collapses each branch map g to the constant at its self-evaluation $g(s)$; its image is A . The data it discards is precisely the non-constant branch maps $g : S \rightarrow T$ — the store-of-store structure that $\triangleright$ carries and $\otimes$ forgets.*

For orientation we also recall how the store comonad’s own comultiplication sits over the retraction; this is the sense in which “ $\otimes$ is the diagonal of $\triangleright$.”

Proposition 4.3 ([5, Thm. 3.1]). *Let $d : S \rightarrow S \times S$ be the diagonal and $\Delta d : \Delta S \rightarrow \Delta(S \times S)$ its store image (shape $s \mapsto (s, s)$, backward π_2). Then $r_{S,S} \circ \delta = \Delta d$: the store comultiplication δ is a lift of the $\otimes$ -diagonal along the retraction r .*

Corollary 4.2 is the crux for the comparison. The workers retract keeps a point-valued datum ($g(s) \in T$, equivalently the entangled state $S \times T$) and discards the branch map’s dependence on inputs other than the current one.

5. THE CONJECTURE, AND WHERE IT BREAKS

We can now line the two keep/discard splits up.

	kept	discarded (tail)
SGF factorization (Prop. 3.3)	$F/1$: right adjoint	Σ : coproduct (left adjoint)
Workers retract (Cor. 4.2)	$A = \Delta(S \times T)$	non-constant branch maps $g : S \rightarrow T$

The parallel is real: in each row a rich construction is split into a piece the relevant correspondence recognises and a tail it does not. It motivates the following.

Conjecture 5.1 (worker = bicomodule/PRA). *There is a choice of polynomial comonads (c, d) for which the BHM grading $B = \Delta S \triangleright \Delta T$ presents a (c, d) -bicomodule, equivalently a prafunctor $c\text{-Set} \rightarrow d\text{-Set}$; under the aggregation factorization of Proposition 3.3, the retract $A = \Delta S \otimes \Delta T$ of Theorem 4.1 is the right-adjoint (bicomodule-content) factor, and the data discarded by the idempotent e of Corollary 4.2 is the coproduct tail Σ . Consequently “worker” = “parametric right adjoint / bicomodule,” and the retract kernel = the aggregation tail.*

We do not claim Conjecture 5.1. Two gaps stand between the suggestive table and a proof; naming them precisely is the real content of this section.

Gap (TODO for a prove session) 5.2 (the apparent reversal of adjointness). *The two “discarded tails” point in opposite adjoint directions, and the conjecture as stated papers over this. In Proposition 3.3 the discarded factor is Σ , a coproduct, hence a left adjoint; the kept factor $F/1$ is the right adjoint. In Corollary 4.2 the discarded data is the set of branch maps $g : S \rightarrow T$. But branch maps are exponential data: they are the elements of T^S , and the power $(-)^S$ is a right*

adjoint (right adjoint to the copower $S \times (-)$) — the adjoint flavour the SGF factorization keeps in its $F/1$ factor. On the face of it, then, the workers retract discards right-adjoint content while the SGF factorization discards left-adjoint content. Either (i) the correct comparison reverses one of the two splits (e.g. via the handedness of Remark 3.4, or by comparing σ rather than e), so that the exponential branch data reappears on the kept side; or (ii) the analogy is only formal and Conjecture 5.1 is false. Deciding which is the first task of a proof session. A concrete sub-question: is the exponential T^S in a B -shape the connected-limit content that $F/1$ retains, with the genuine Σ -tail located elsewhere (e.g. in the coproduct $S \cdot (-)$ that makes $\Delta S = S \cdot y^S$ a sum of representables in the first place)?

Gap (TODO for a prove session) 5.3 (which two comonads?). Proposition 3.2 is about maps between fixed comonads, $\text{Comod}(\mathbf{Poly})(c, d)$, whereas $\Delta S \otimes \Delta T$ and $\Delta S \triangleright \Delta T$ are monoidal products of two comonads, producing new polynomials rather than a map between two others. Before the correspondence can bite, one must say what plays the role of (c, d) . Candidates: (a) take $c = \Delta T$, $d = \Delta S$ and read $\Delta S \triangleright \Delta T$ as (the carrier of) a bicomodule via the composite comonad structure — but $\triangleright$ of two comonads need not be a comonad, so this requires care; (b) fix $c = d = y$ (the terminal category) and read everything as an endo-prafunctor of $\mathbf{Set}$, in which case the correspondence degenerates and the comparison is only with the single-variable Proposition 3.3; (c) use the store comonads’ status as codiscrete categories (Definition 2.4) to make $\Delta S, \Delta T$ the two objects and the graded composition a genuine bicomodule between them. Only (c) has a chance of engaging the full Theorem 3.1; verifying that the BHM grading is the corresponding bicomodule is not done here.

Remark 5.4 (the nearest neighbour, and why it is not a shortcut). The graded-comonad reading of workers has an established relative: Capucci–Myers’s “contextads” [2], where a polynomial monad transposes to a dependently graded comonad and composition multiplies the grade. Its Theorem A.4 ($\text{Kl}(T) \simeq \text{Ctx}(\odot)$ for a parametric right adjoint monad T) is stated for exactly the PRA class that appears here — but it is left *unproven* in [2]. So the literature already gestures at “graded composition lives in the PRA world,” but it does not supply the bicomodule identification we need, and cannot be cited as a proof of Conjecture 5.1. If anything, settling Gaps 5.2 and 5.3 would bear on Theorem A.4, not the other way round.

6. WHAT A PROOF SESSION MUST SETTLE

The correspondence of Spivak–Garner–Fairbanks and the workers retract are each solid; the bridge between them is not yet built. This note has reduced “is a worker a bicomodule?” to two precise tasks.

- (1) **Resolve the adjointness reversal (Gap 5.2).** The exponential branch data T^S that the retract discards is right-adjoint-flavoured, whereas the SGF factorization discards a coproduct. Determine whether a handedness or σ -versus- e change of comparison puts the two splits on the same side, or whether the analogy fails.
- (2) **Pin the comonads (Gap 5.3).** Exhibit the pair (c, d) for which $\Delta S \triangleright \Delta T$ is a (c, d) -bicomodule, presumably using the store comonads as codiscrete categories, and check that the corresponding prafunctor’s Proposition 3.3 factorization has the workers retract as its right-adjoint factor.

If both succeed, Conjecture 5.1 holds and “worker” acquires a clean semantics inside $\mathbf{Poly}$: a worker is a parametric right adjoint, its composition is bicomodule composition, and the retract that separates entanglement from nesting is the aggregation factorization. That is worth having — it would place the state axis of the compositional-correctness program inside the same framed bicategory $\mathbf{Cat}^\sharp$ that already houses the directed (retrofunctor) axis. If either fails, the failure is

itself informative: it would say that state-entanglement is a *different* kind of collapse from data-aggregation, and would sharpen the question of what the store-of-store structure really is. Both outcomes are progress; neither is available without the proof session.

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